

Generalised Linear Models

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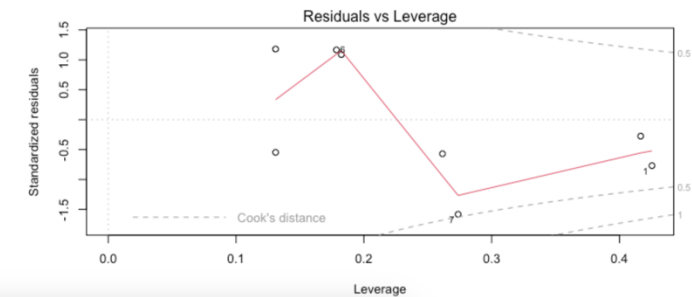
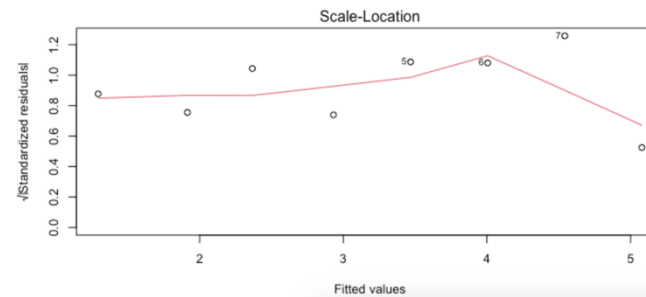
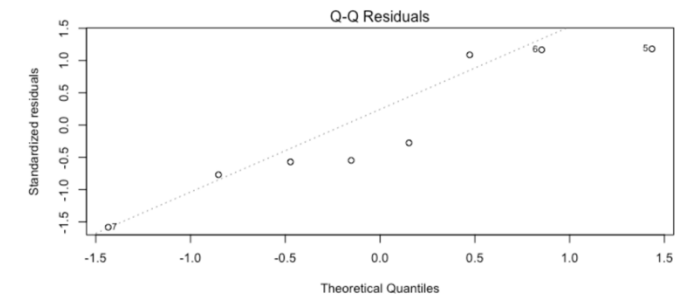
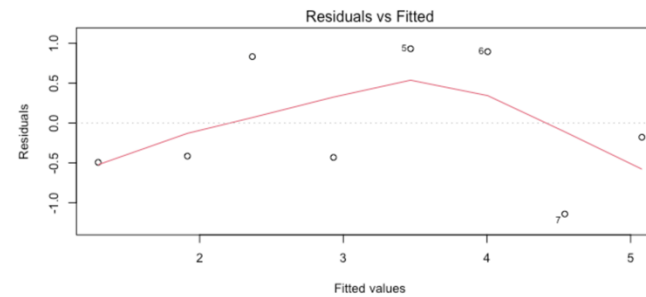
Intended Learning Outcomes

You will be able to:

- Explain and validate linear models by interpreting diagnostic plots
- Differentiate between a linear model and generalised linear models of binomial and poisson data
- Explain why generalized linear models are preferential over simple linear models
- Outline the three steps of generalized linear model

Assumptions of Linear Models

- Defined response/dependent variable
 - Different from correlation
 - Causation
- Independent variables are not collinear
- Relationship is linear
 - Examined through plotting
- Residuals are normally distributed
- Homogeneity of variances
- No outliers



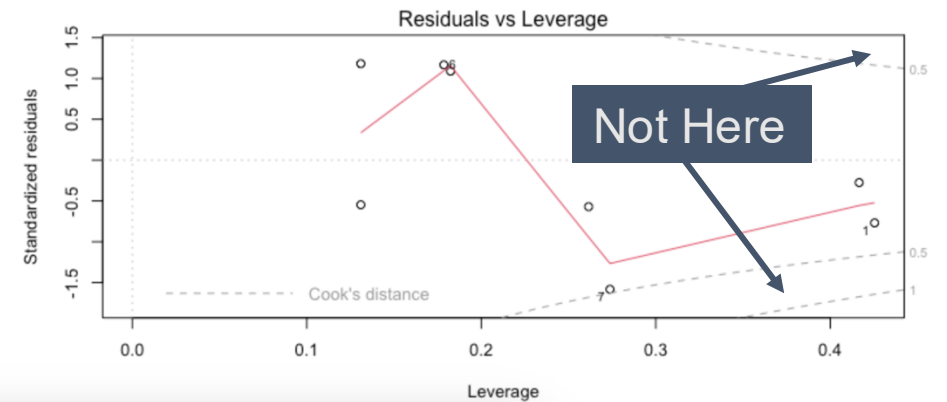
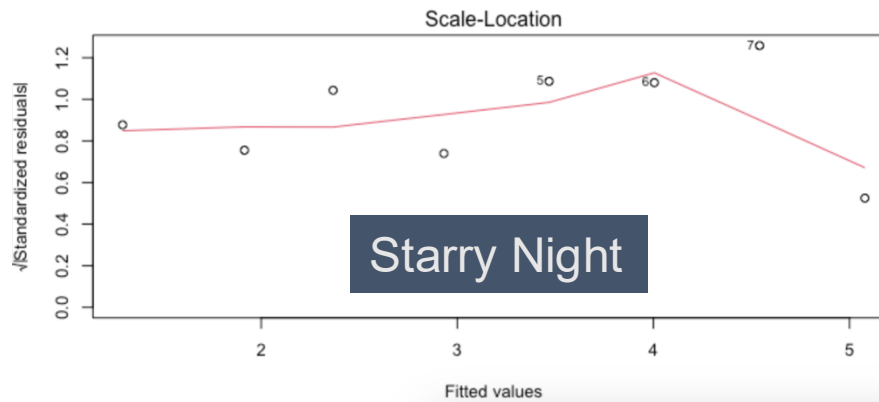
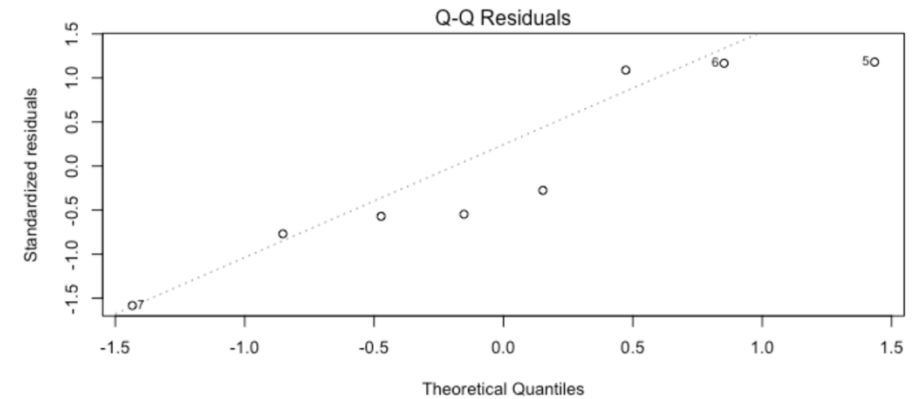
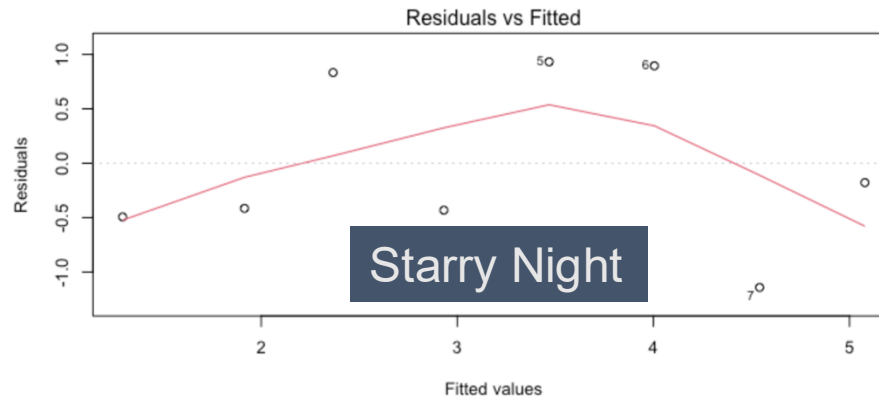
Assumptions of Linear Models

(1) & (3)
Homogeneity of
Variances

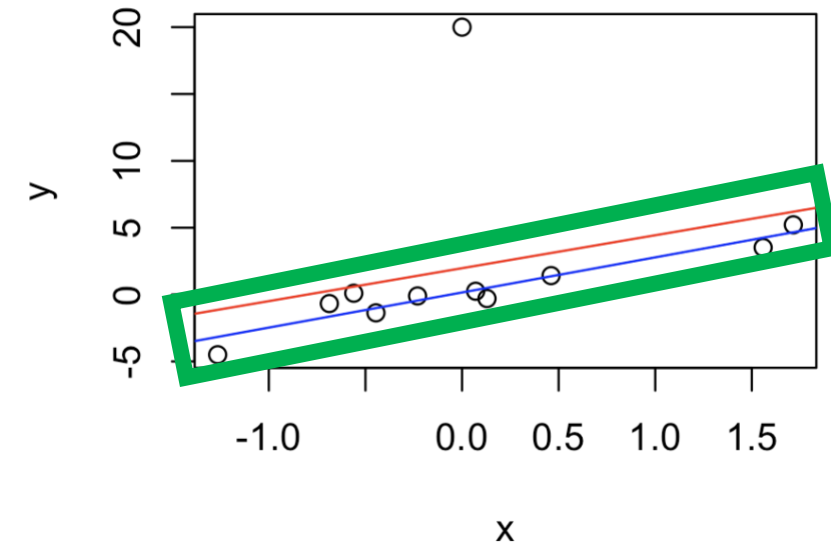
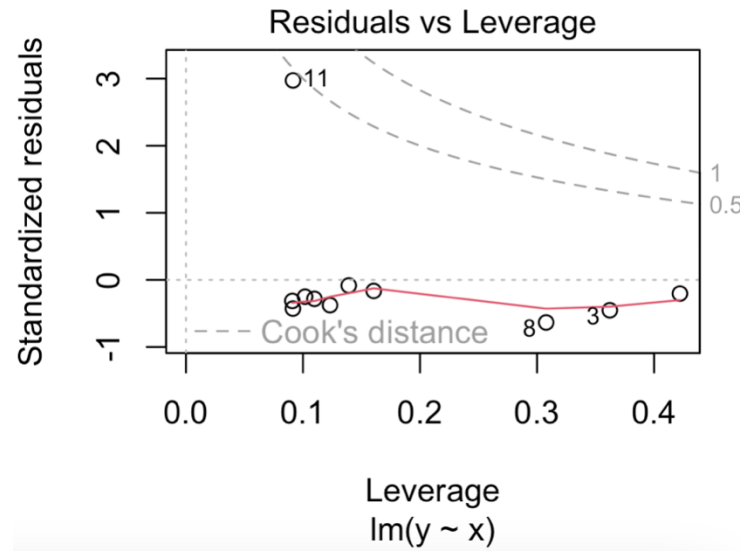
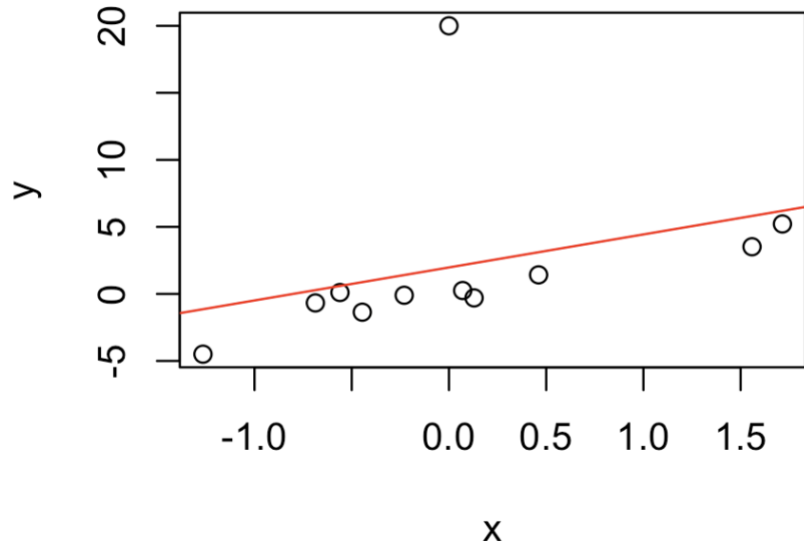
$$\varepsilon_i \sim \gamma_i$$

(2) Normality of
the residuals

(4) Outliers
 $\varepsilon_i \sim \text{leverage}$



What is leverage?



LMs vs GLMS

- Response variable data type
 - Unconstrained vs Constrained
- Model fitting approach
 - Ordinary Least Squares vs Maximum Likelihood
- Assumptions
 - Means and variance

Linear Models

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

The diagram illustrates the components of the linear model equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. Arrows point from the following labels to their respective terms in the equation:

- Response Variable** points to y_i .
- Intercept** points to β_0 .
- Slope of Explanatory Variable** points to β_1 .
- Error Term** points to ε_i .

Linear Models

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

The diagram illustrates the components of the linear model equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. Arrows point from descriptive labels to the corresponding terms in the equation:

- An arrow points from the label "Response Variable" (enclosed in a red box) to y_i .
- An arrow points from the label "Intercept" to β_0 .
- An arrow points from the label "Slope of Explanatory Variable" to β_1 .
- An arrow points from the label "Error Term" to ε_i .

Linear Models

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

The diagram illustrates the components of the linear model equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. Arrows point from descriptive labels to each term in the equation: y_i is the Response Variable, β_0 is the Intercept, β_1 is the Slope of Explanatory Variable, and ε_i is the Error Term. The labels 'Response Variable' and 'Error Term' are enclosed in red boxes.

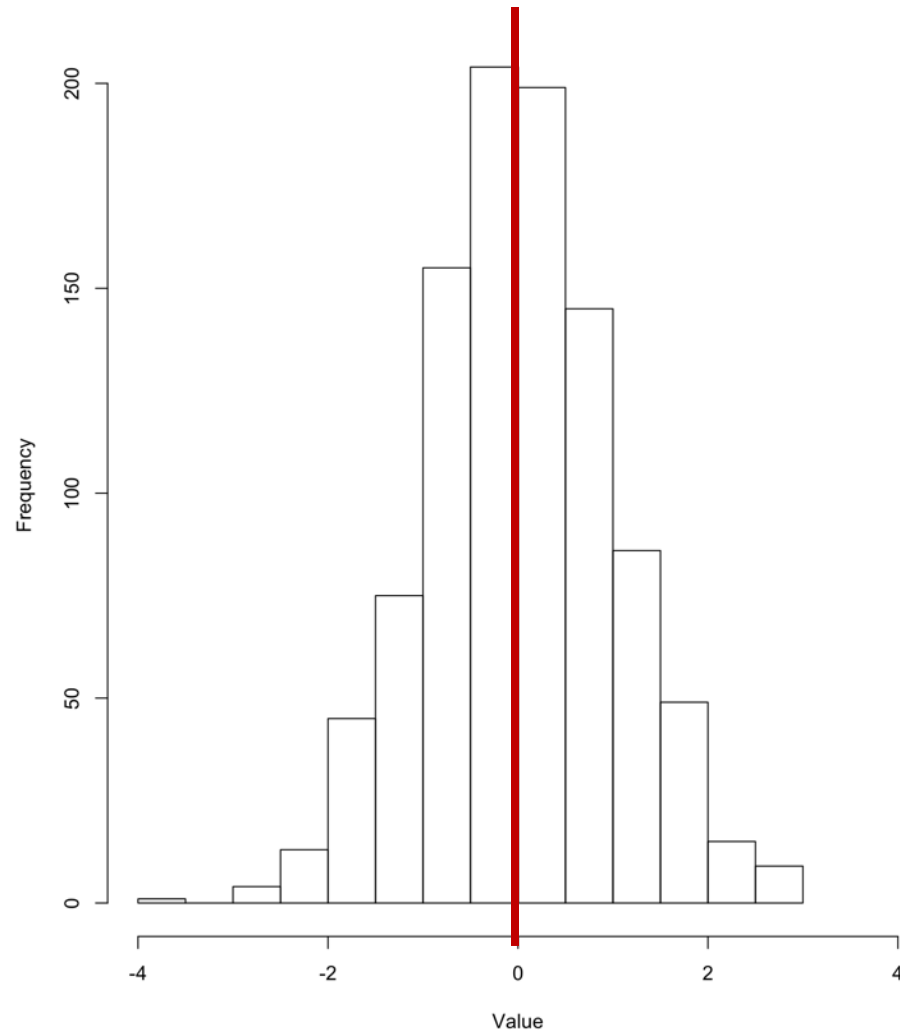
Response Variable

Intercept

Slope of Explanatory Variable

Error Term

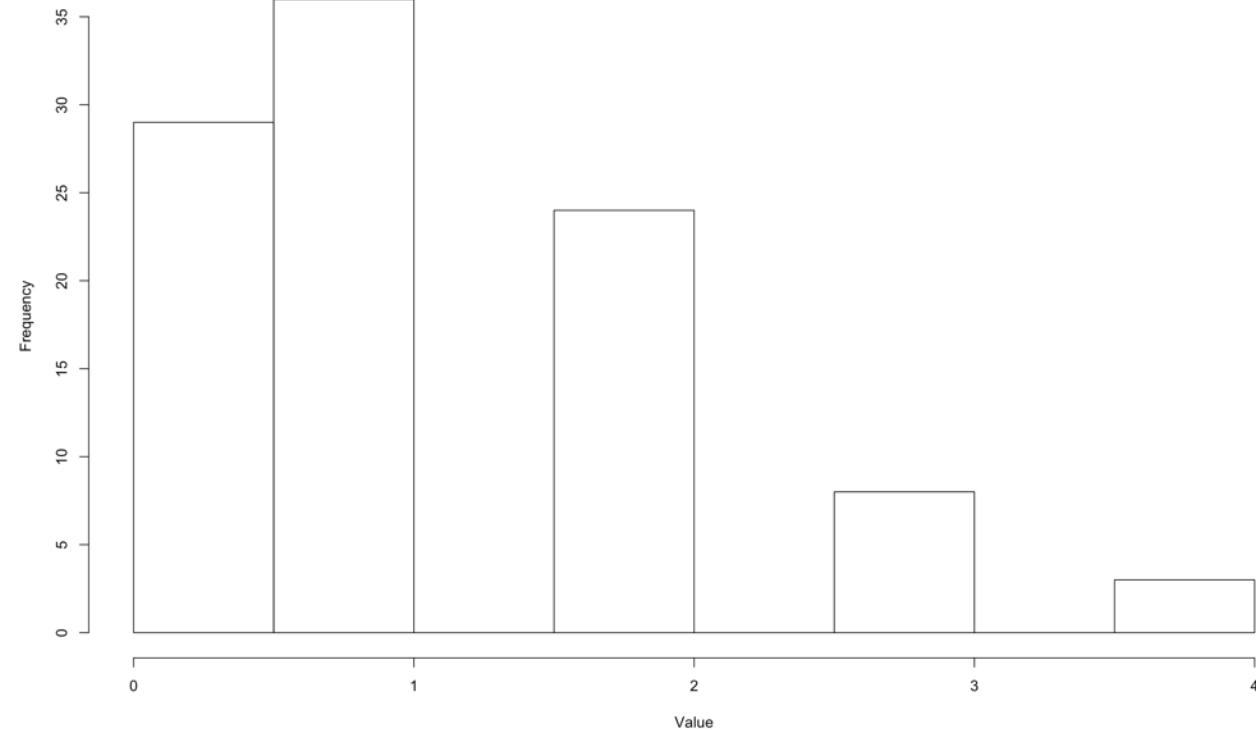
Revisiting Data Distributions- Normal



- -Infinity to Infinity (technically)
- Mean represents the centre of the data
- Examples:
 - Leaf area
 - Plant height
 - Biomass
 - Feeding rate

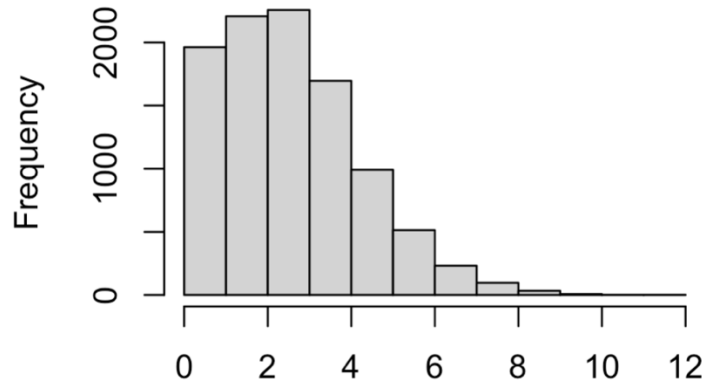
Revisiting Data Distributions- Poisson

- Count Data
- **Constrained** to absolute whole numbers
- Typically right skewed
- Examples:
 - Number of Species
 - Number of Enzymes
 - Heartbeat
 - Number of Offspring
- Mean is equal to variance??

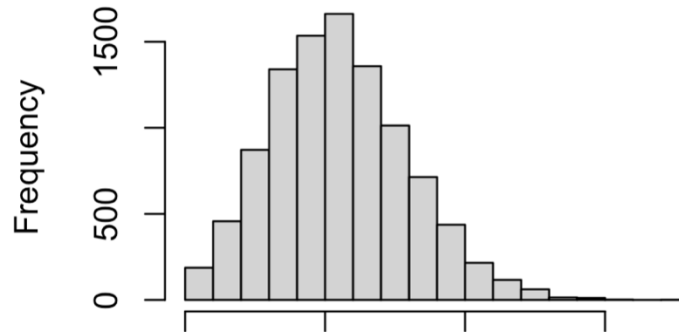


Mean is equal to variance

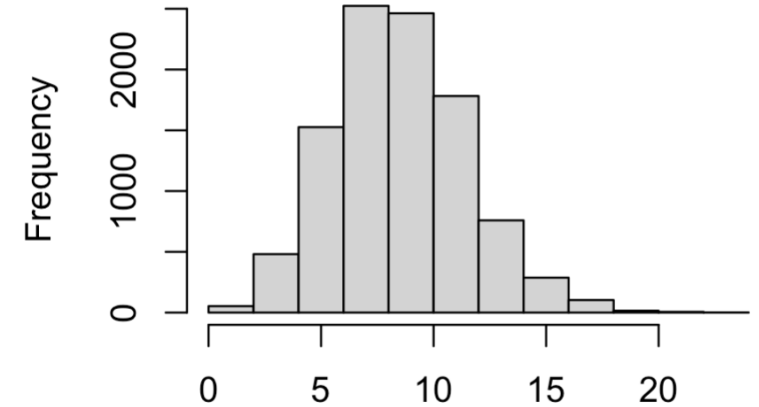
Mean is 3



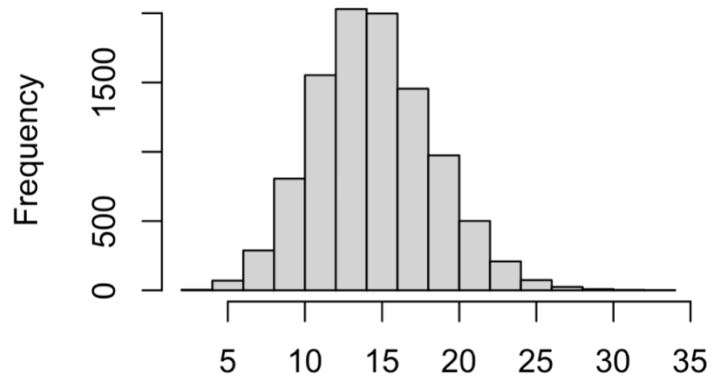
Mean is 6



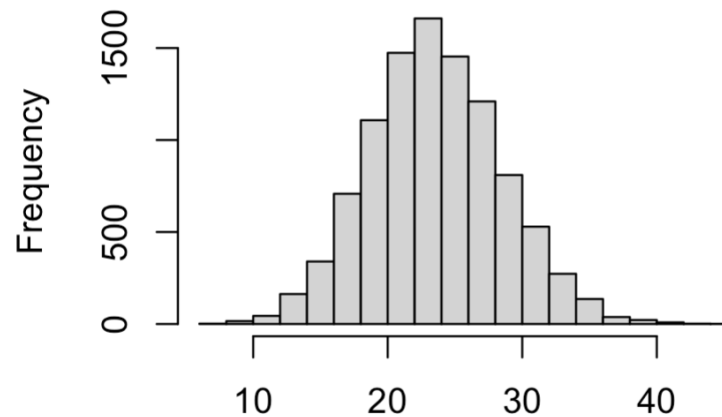
Mean is 9



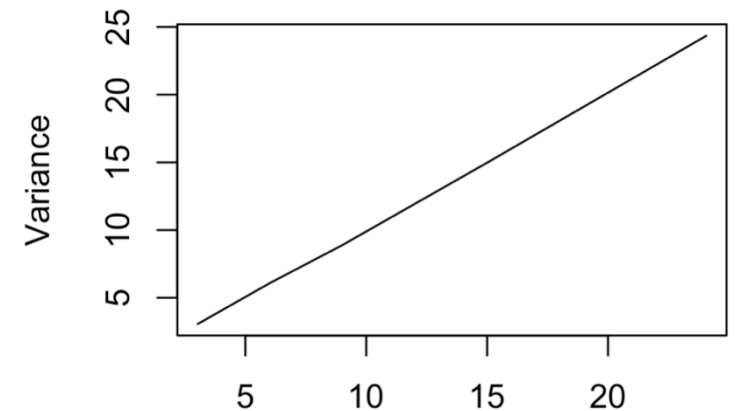
Mean is 15



Mean is 24



Species Richness



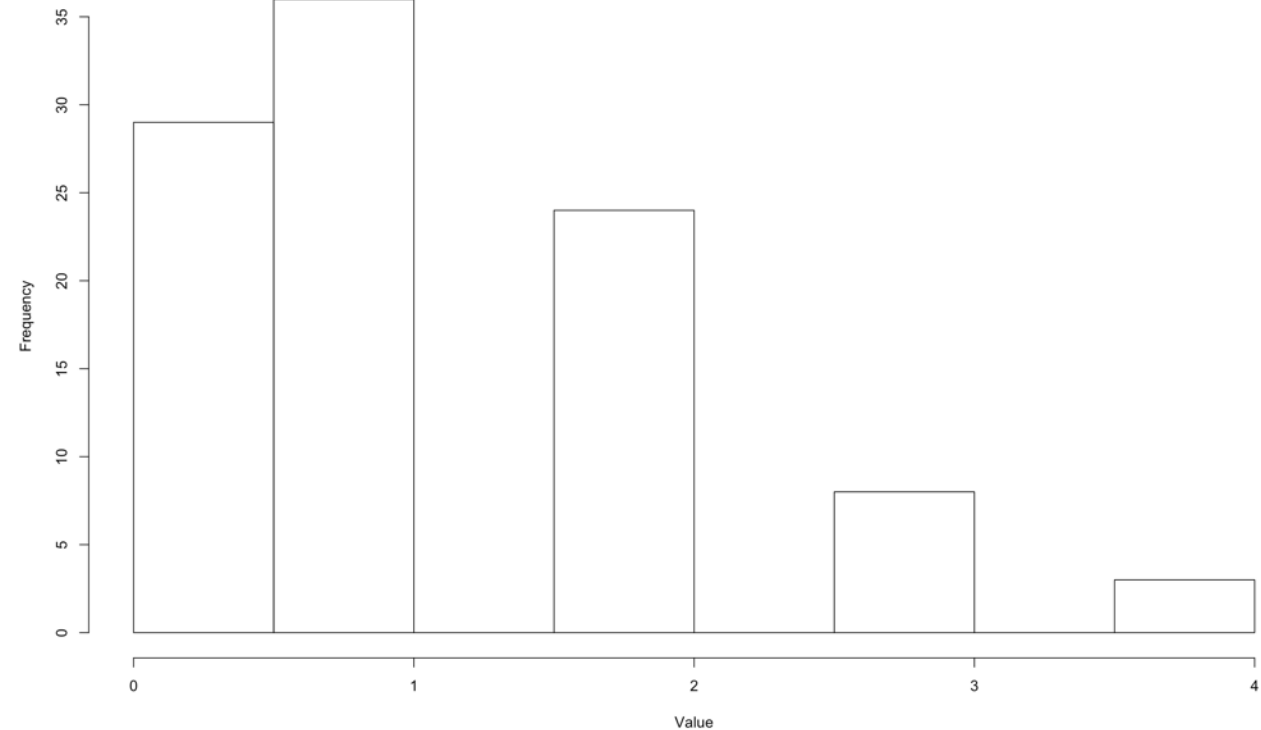
Species Richness

Species Richness

Mean

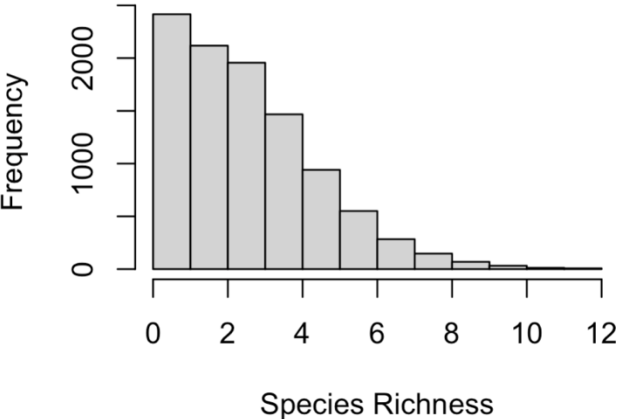
Revisiting Data Distributions- Negative Binomial

- Count Data
- **Constrained** to absolute whole numbers
- Typically right skewed
- Examples:
 - Number of Species
 - Number of Enzymes
 - Heartbeat
 - Number of Offspring
- Mean is **not** equal to variance

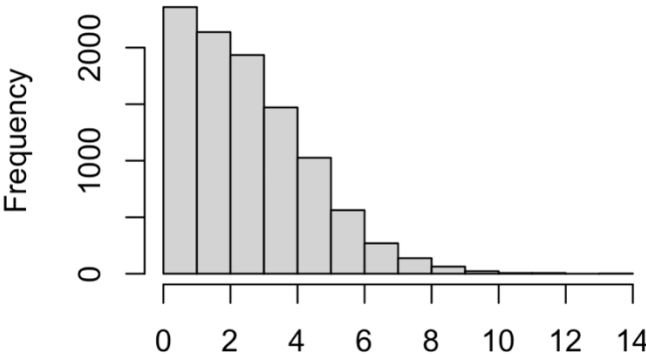


Revisiting Data Distributions- Negative Binomial

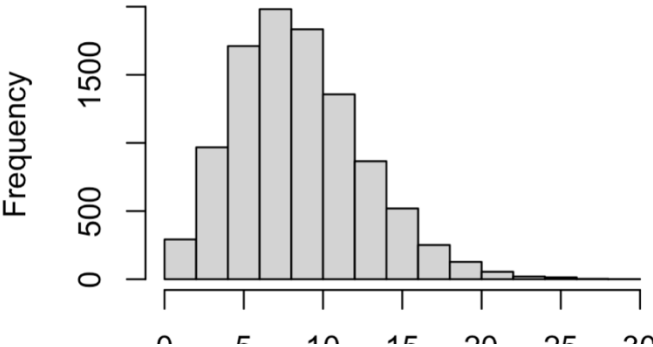
Mean is 3



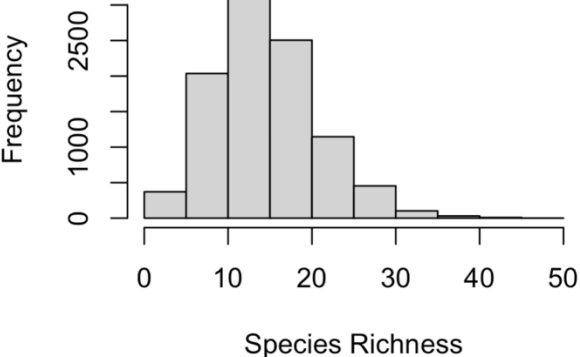
Mean is 6



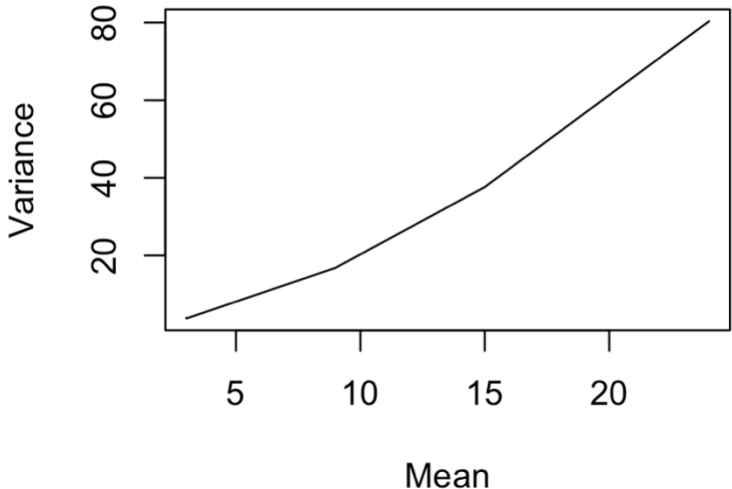
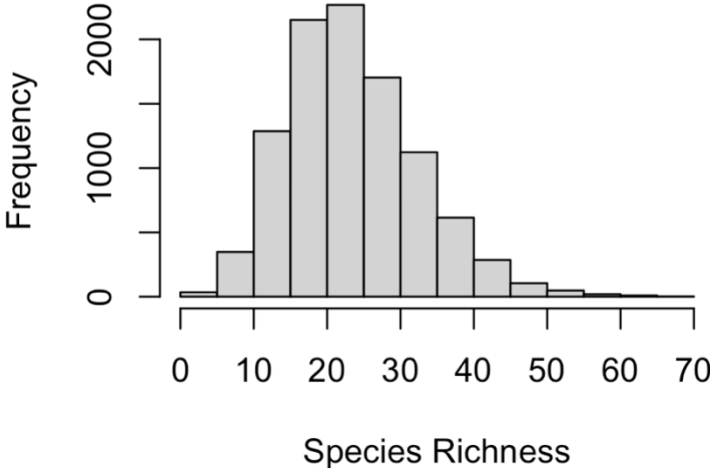
Mean is 9



Mean is 15

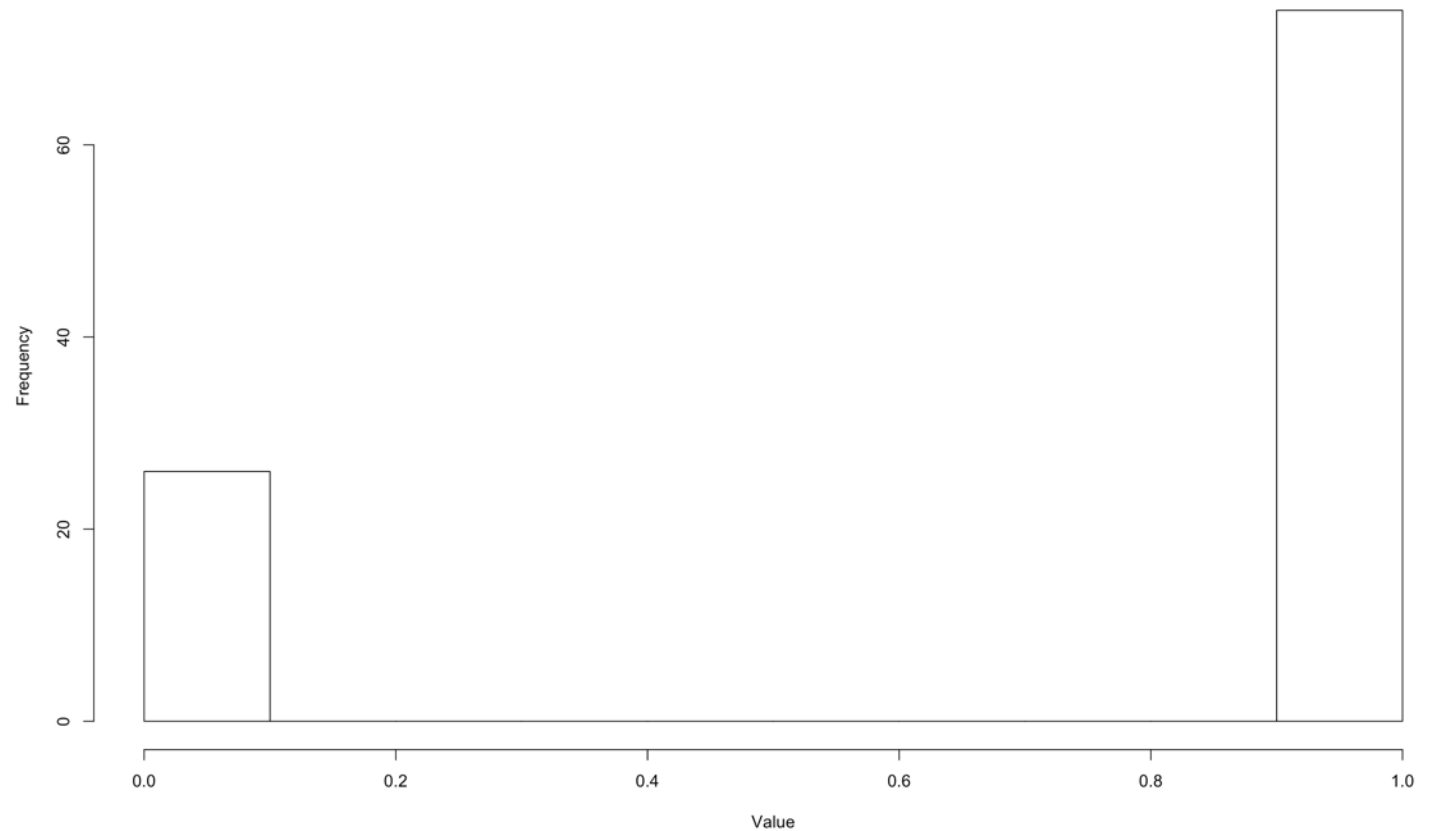


Mean is 24



Revisiting Data Distributions- Binomial

- **Constrained** between 0 and 1
- Examples:
 - Proportions
 - Presence/Absence Data



LMs vs GLMS

- Response variable data type
 - Unconstrained vs Constrained
- Model fitting approach
 - Ordinary Least Squares vs Maximum Likelihood Estimation
- Assumptions
 - Means and variance

OLS vs MLE

- Model fitting approach
 - Ordinary Least Squares vs Maximum Likelihood Estimation



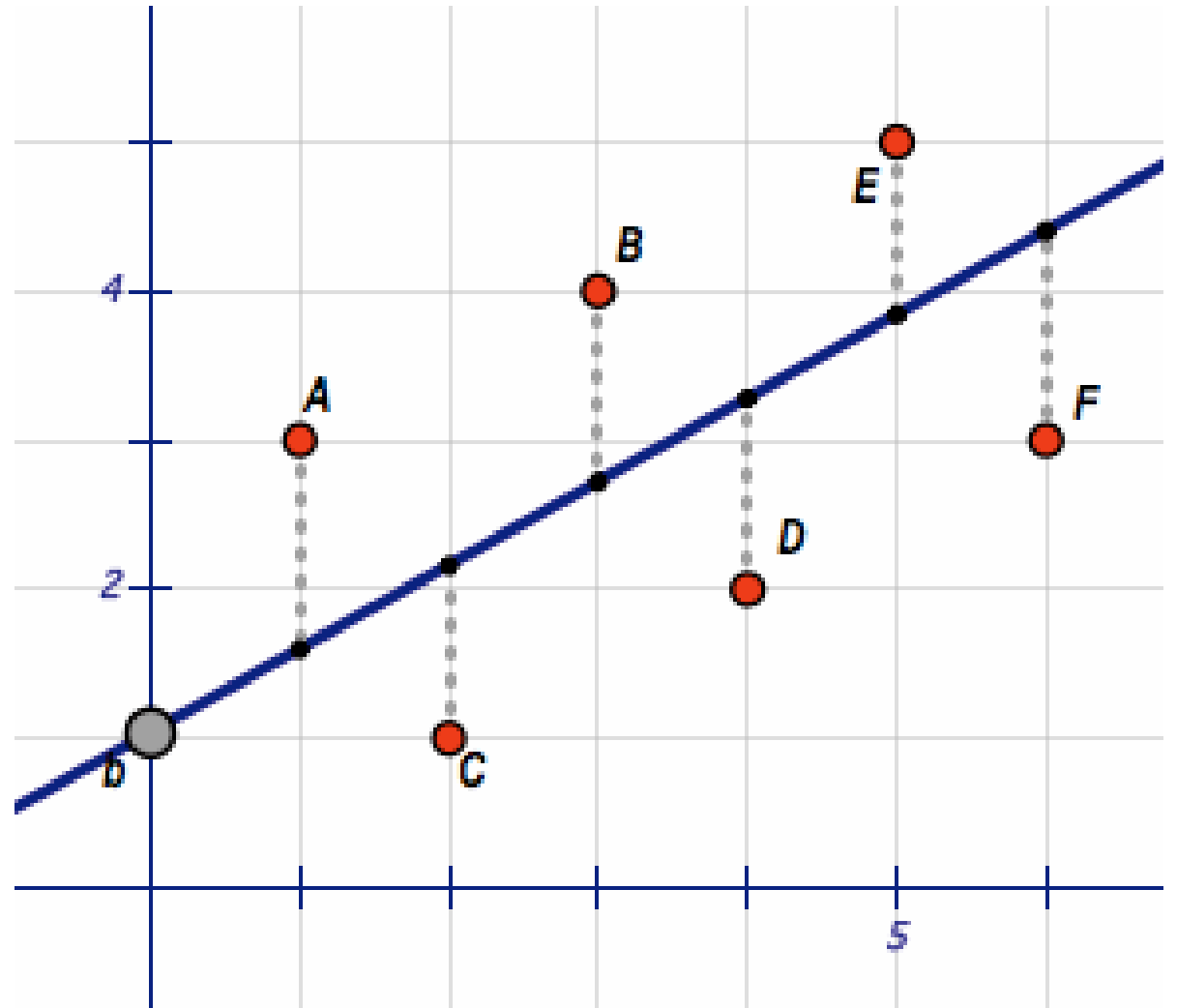
Fits a line that
minimizes the
residual sum
of squares



Fits a line that
maximises the
log-likelihood

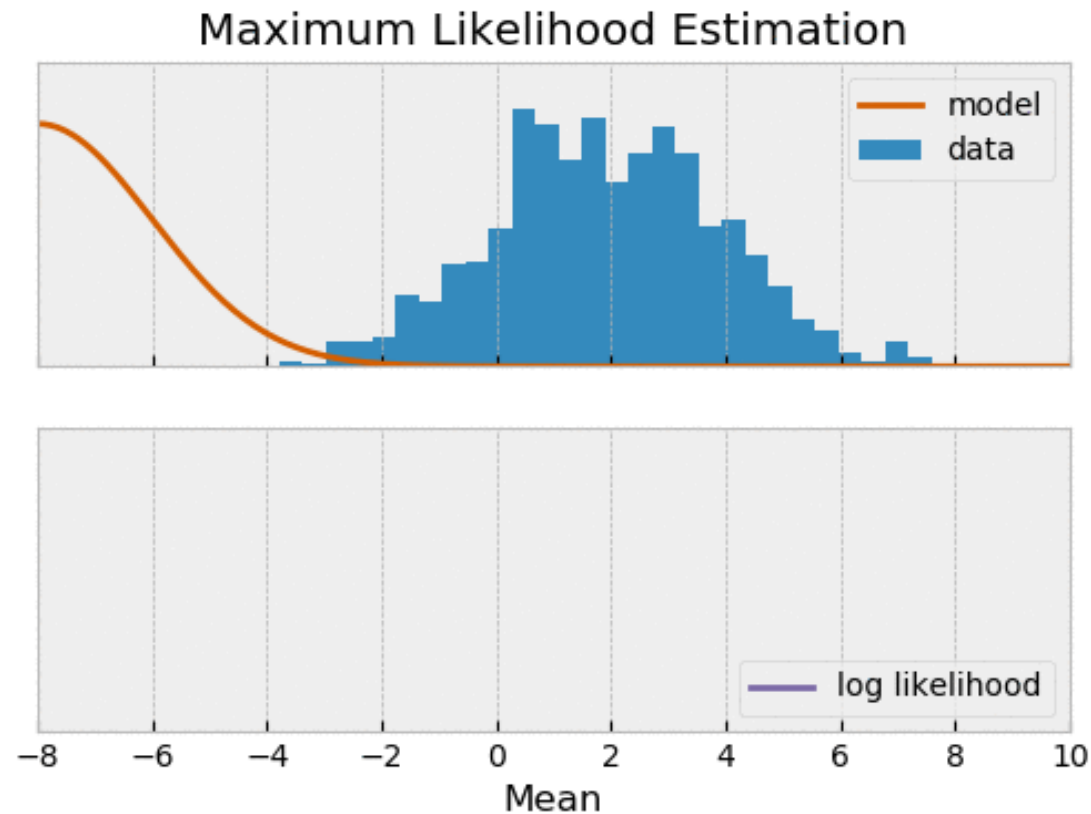
OLS vs MLE

- Model fitting approach
 - Ordinary Least Squares



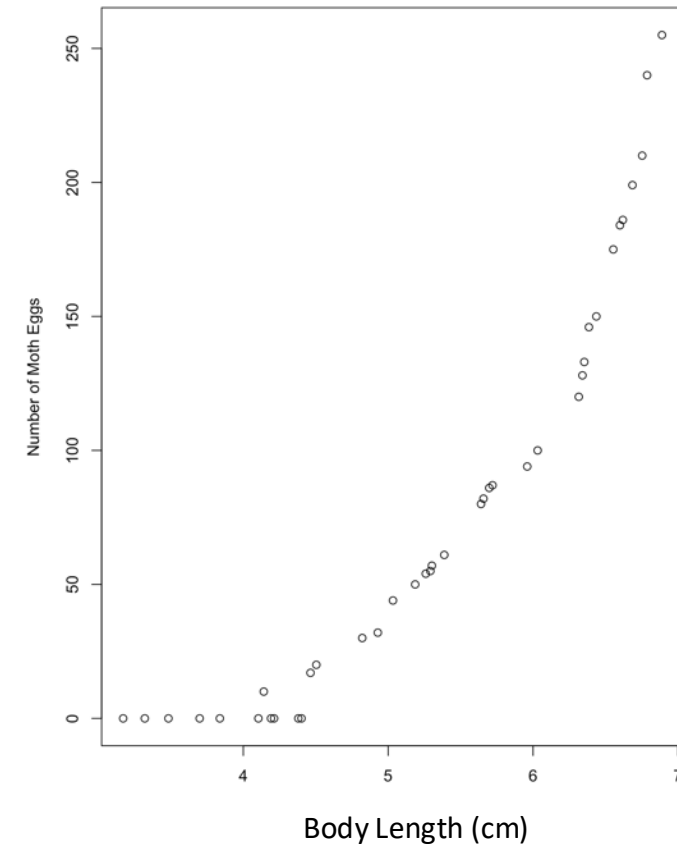
OLS vs MLE

- Model fitting approach
 - Maximum Likelihood Estimation

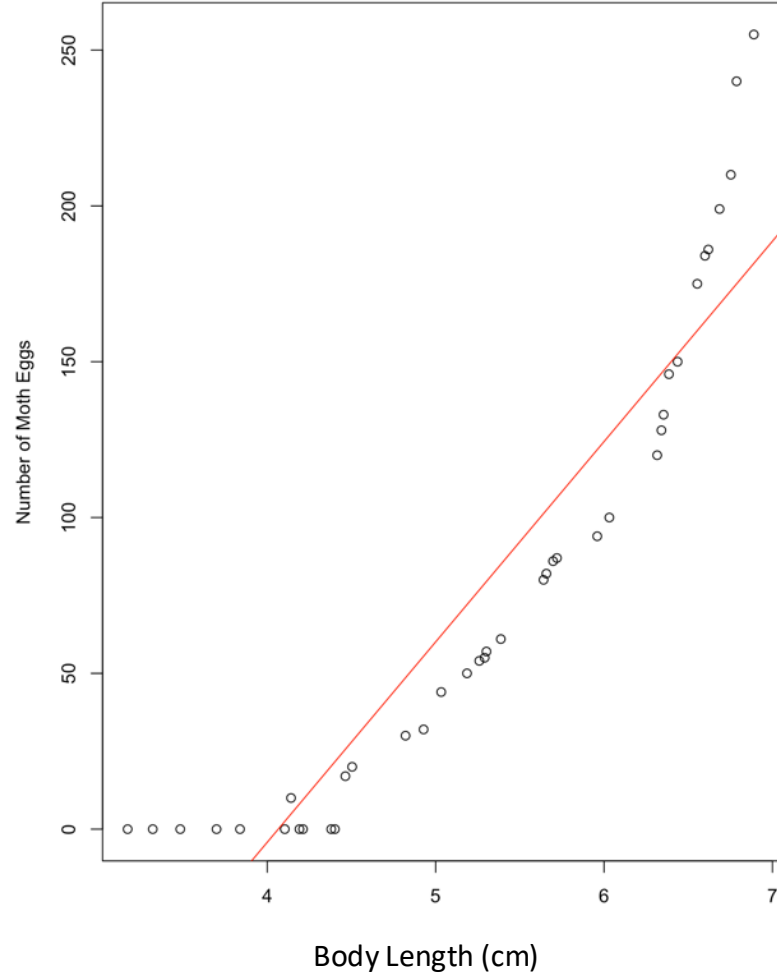


Example

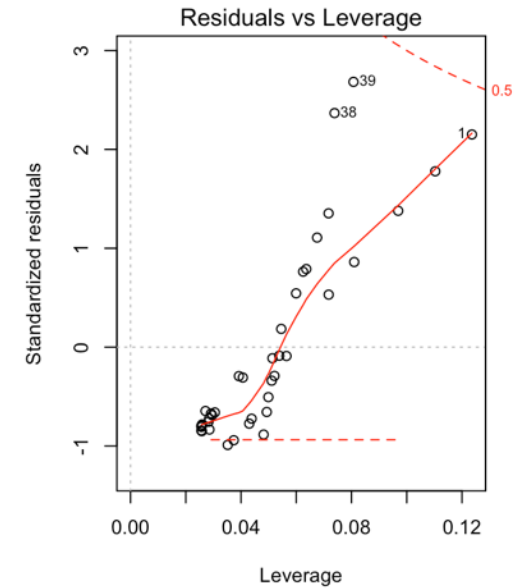
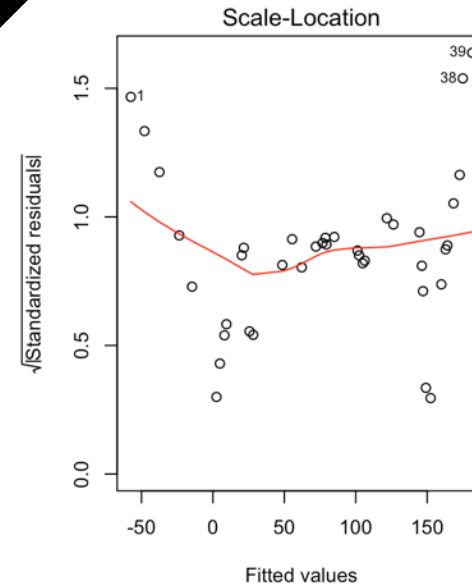
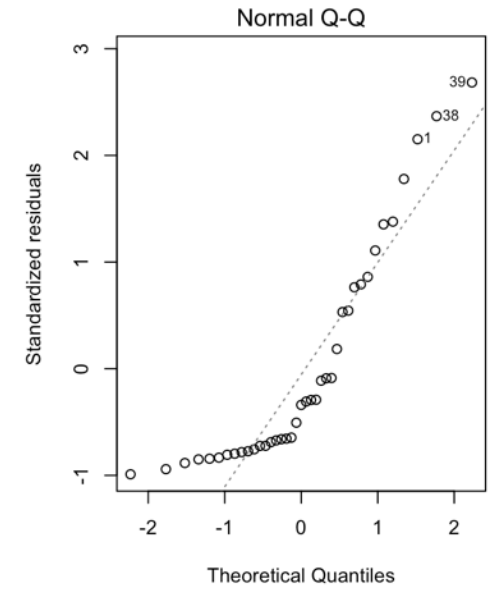
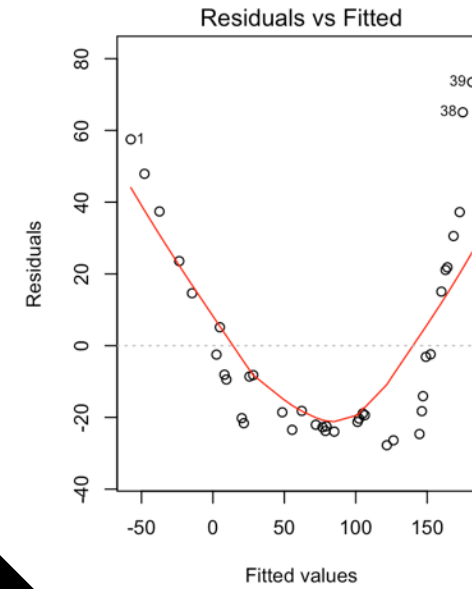
- Number of eggs laid and female body length of vapourer moth



Example

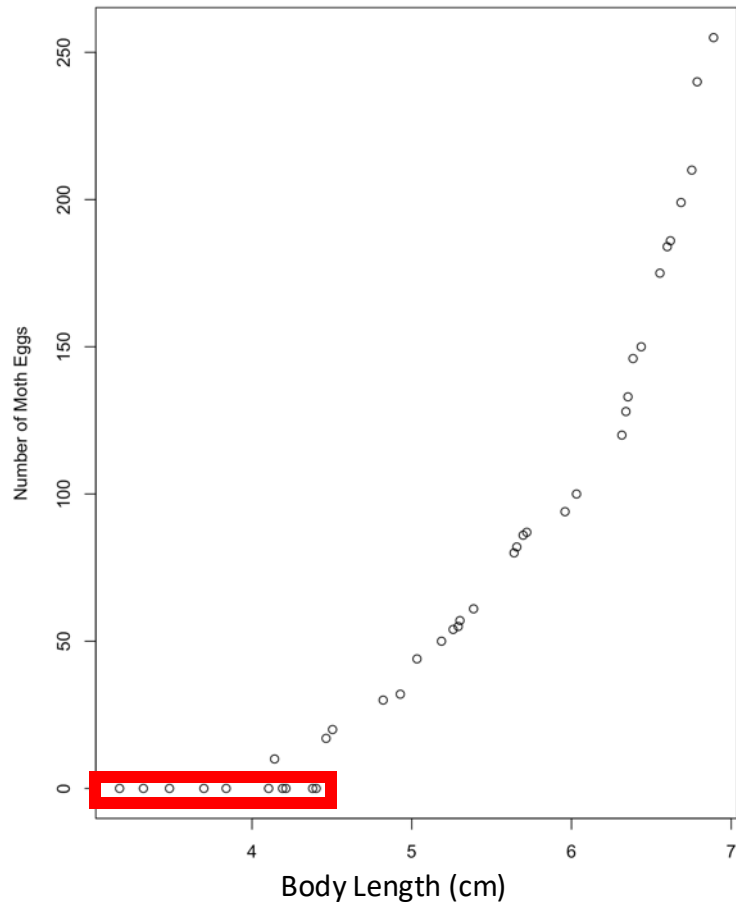


Diagnostics

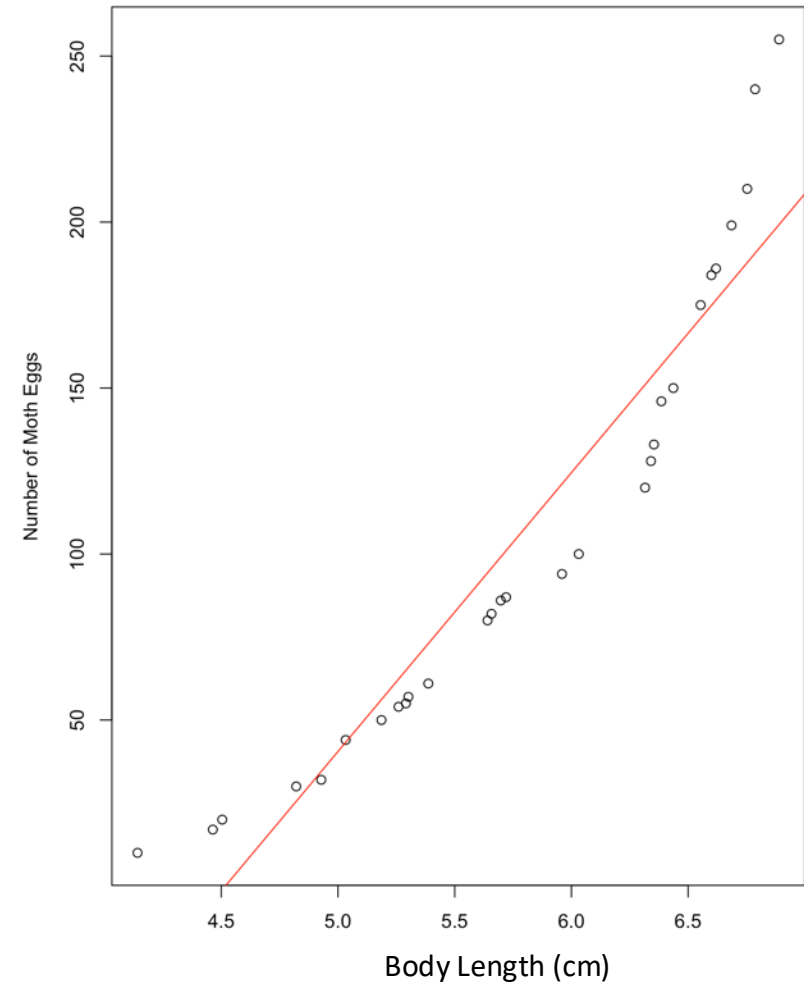


Solutions?

- Log transformations of count data



**Lose ~26% of
Data**



Generalised Linear Models

- Any GLM consists of three steps:
 1. Choosing a distribution for the response variable
 2. Specifying the linear function of covariates and/or fixed factors
 3. Choosing a link between the predictor function and the mean of the distribution

Generalised Linear Models

1. Distribution of response = normal
2. Predictor function = $\beta_0 + \beta_1 x_i + \varepsilon_i$
3. Link between the predictor and the mean of the distribution:

$$\begin{array}{ccc} y_i = & \beta_0 + \beta_1 x_i + \varepsilon_i \\ \uparrow & \uparrow \\ \text{Predicted} & \text{Linear predictor} \\ \text{Response} & \end{array}$$

Link Functions

- Generalised linear model relates the response distribution to the linear predictor via a link function
- There are different link functions for different distributions

The diagram illustrates the equation $h(y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i$. A horizontal line with a downward arrow on the left points to the term $h(y_i)$, which is labeled "Predicted Response". A red underline is placed under the expression $\beta_0 + \beta_1 x_i + \varepsilon_i$, with an upward arrow pointing to it labeled "Linear predictor". A long horizontal line extends from the right side of the equation, with a downward arrow pointing to the label "Link function".

$$h(y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Link function

Predicted Response

Linear predictor

Link Functions

Normal

Leaf Mass

Bill Width

Height

Weight

Poisson (Counts)

Number of Species

Number of Enzymes

Number of Offspring

0, 1, 2, 3, 7

Binomial

0,1,1,0

Female, Male, Female, Male

Absent, Present, Present, Absent

Heads, Heads, Heads, Tails

Identity

**Log-linear
(natural)**

Logit

Summary

- Generalised linear models estimate the linear estimates via maximum likelihood estimation and are able to handle constrained data types
- The response variable is related to the linear predictor via a link function and these are specific to distribution families
- We first need to establish the distribution of our response variable and the linear predictor. With both of these, we can select an appropriate link function